Do Now · retrieval

Interrupts and service routines

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Define fetch-decode-execute.

Q2

What causes a process to enter Waiting?

Q3

What does an OS scheduler choose?

Q4

Simplify $(A \wedge B) \vee (A \wedge \neg B)$, showing factoring.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — An interrupt signals an event such as device completion to the processor.

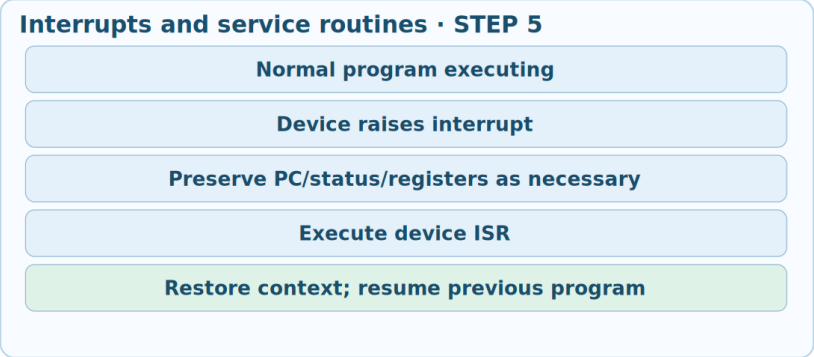
Core idea 2

Trace — The CPU saves sufficient execution context before servicing the event.

Core idea 3

Explain — An ISR handles the interrupt, acknowledges it, then restores normal execution state.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

interrupt — CPU attention request

ISR — Interrupt Service Routine

context — State for safe resume

polling — Repeatedly checking event state

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

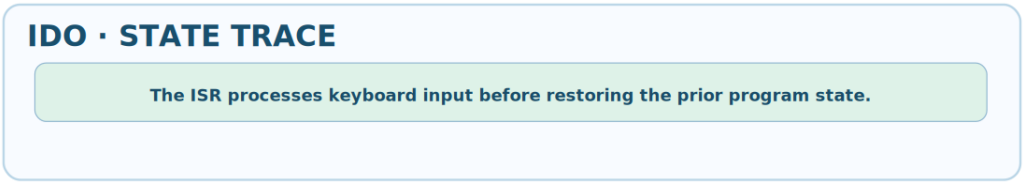
Problem

Keyboard triggers an interrupt during a text editor. Trace the stages.

Follow the worked method

1. Predict first — The ISR processes keyboard input before restoring the prior program state.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
The ISR can acknowledge completion and allow queued device work to proceed; the interrupted task resumes.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define interrupt, ISR and one source of interrupts. [4]

Question 2

Why does CPU context need to be saved/restored around a device ISR? [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare polling and interrupt-driven I/O for infrequent device input. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An interrupt handler can overwrite original registers without preserving any state.

Use key vocabulary from memory

1. Explain interrupt in your own words.

2. Explain ISR in your own words.

3. Explain context in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.